AWS CLOUD ENGINEERING

AWS Secure VPC Architecture

Public & Private Networking using **IGW**, **NAT Gateway** & **Bastion Host**

A hands-on walkthrough of designing and deploying a production-grade, secure AWS network — from CIDR planning to validated connectivity.

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PROBLEM STATEMENT

Why This Architecture?

Default AWS setups leave resources exposed. A production-grade VPC must enforce isolation, controlled access, and minimal attack surface.

Direct Internet Exposure

Public cloud resources are often reachable from the internet without any intermediary protection.

Backend Isolation Required

Databases and app servers must be shielded in private subnets — no public IPs, no direct ingress.

Controlled Access Needed

Open SSH to private instances is a security risk. A bastion host enforces a single, auditable entry point.

Production-Style Goal

Build a secure, scalable AWS network that mirrors real-world enterprise architecture patterns.

Why Not Directly Expose Backend Systems?

- **Direct exposure increases attack surface** — every instance becomes a potential entry point
- **Violates least privilege principle** — only necessary services should be internet-accessible
- **Not suitable for production environments** — enterprise standards require isolation and controlled access

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ARCHITECTURE OVERVIEW

VPC Design

Core components and separation of concerns for a secure AWS network architecture.

Core Components

- **Custom VPC** — CIDR planned for subnet isolation
- **Public Subnet** — Internet-facing resources only
- **Private Subnet** — Secure backend systems, no public IPs
- **Internet Gateway** — Bidirectional external connectivity
- **NAT Gateway** — Outbound-only internet for private subnet
- **Bastion Host** — Single secure SSH jump point
- **EC2 Instances** — Deployed across both subnet tiers

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ARCHITECTURE FLOW

Traffic Flow — End to End

🌐 User Flow (Public Access)

User → Internet → Internet Gateway (IGW) → Web Server EC2 (Public Subnet)

- Only HTTP/HTTPS traffic allowed
- Bastion Host is not part of user flow

🔐 Admin / Developer Access

Admin Laptop → Bastion Host → Private EC2

- SSH (Port 22)
- Access restricted via Security Group (Admin IP only)
- Bastion acts as secure jump server

🔒 Private EC2 Internet Access

Private EC2 → NAT Gateway → Internet

- Outbound only
- Used for updates, package installs, API calls

Private EC2 cannot be accessed directly from the internet — all access is controlled and routed. This architecture follows a production-grade design where public access, administrative access, and private resource access are clearly separated.

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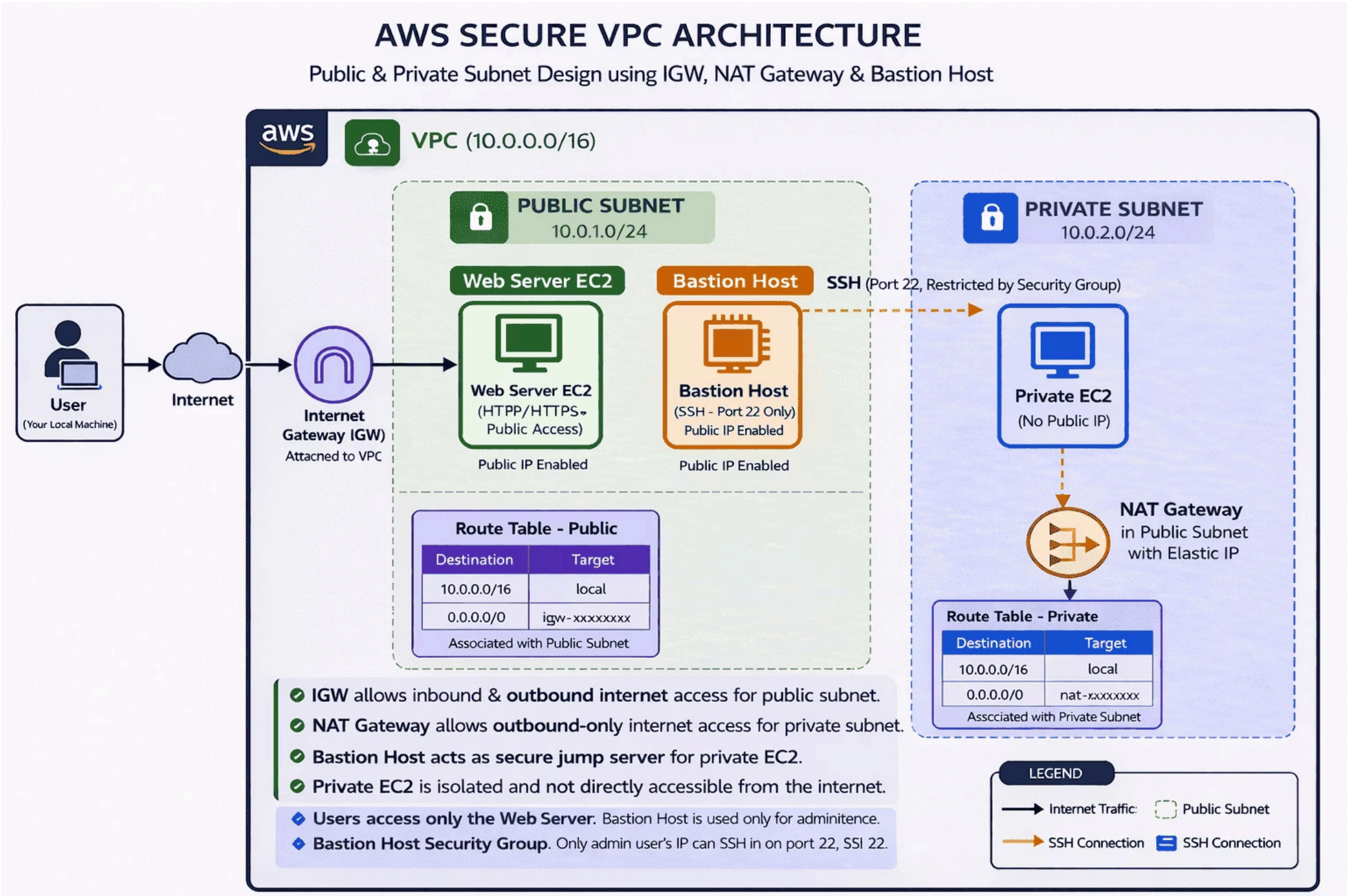
Architecture Diagram



- Users access only the Web Server (not Bastion)
- Bastion Host is used only for administrative SSH access
- Private EC2 has no public IP and is not directly accessible

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IMPLEMENTATION

Public Subnet Setup

Configuration Steps

- Enabled **auto-assign public IP** on subnet
- Attached **Internet Gateway** to VPC
- Route table: `0.0.0.0/0 → IGW`
- Launched EC2 instance with public IP
- Verified web server reachable from internet

> The public subnet route table is the critical control point — it determines what traffic reaches the IGW.

Route Table — Public

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-xxxxxxxx

The route table is the critical control point — it determines what traffic reaches the IGW and what stays internal.

Subnet Settings

- CIDR: `10.0.1.0/24`
- Auto-assign public IP: **Enabled**
- IGW attached: **Yes**

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IMPLEMENTATION

Private Subnet & NAT Gateway

Private Subnet

- Created subnet **without** auto-assign public IP
- Route table points to NAT Gateway
- EC2 instances receive **private IPs only**
- No inbound internet route exists

NAT Gateway

- Deployed in **public subnet**
- Associated with an **Elastic IP**
- Route: `0.0.0.0/0 → nat-xxxxxxxx`
- Enables **outbound-only** internet access

Route Table — Private

```
Destination    Target
-----------    ------
10.0.0.0/16    local
0.0.0.0/0      nat-xxxxxxxx
```

NAT Gateway resides in the public subnet, bridging private instances to the internet for updates and external API calls — without exposing them.

Key Distinction:

IGW = Bidirectional internet access (public subnet) **NAT Gateway** = Outbound-only internet access (private subnet)

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SECURITY

Secure Access via Bastion Host

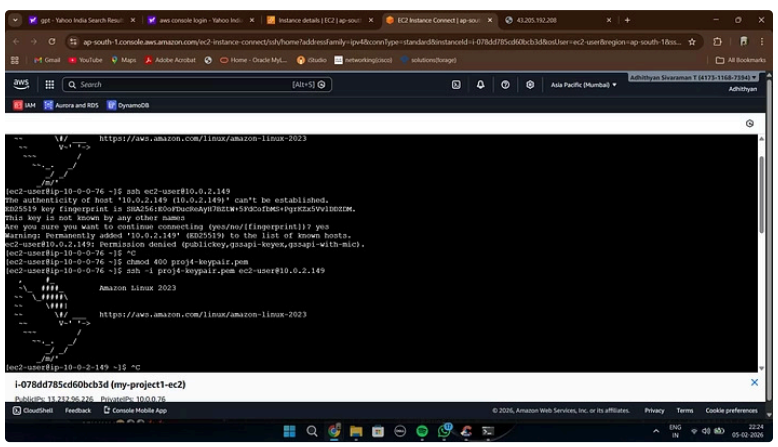
The Bastion Host is the **single controlled entry point** to the private network. Direct SSH to private instances is blocked by security groups and route tables.

1	2	3
Local Machine	**Bastion Host**	**Private EC2**
Initiate SSH with private key	Public EC2 in public subnet — jump server	Accessed via private IP only — no public route

Why Bastion?

- **Bastion Host Security Group:** Allow SSH (22) only from Admin public IP
- **Private EC2 Security Group:** Allow SSH (22) only from Bastion Host Security Group
- Single point of audit — all administrative access is logged
- Reduces attack surface — private instances have no public IPs
- Enforces least privilege — only bastion can SSH to private instances
- Production standard — used in enterprise AWS deployments

This enforces the bastion access pattern and least privilege principle. This architecture follows a production-grade design where public access, administrative access, and private resource access are clearly separated.



SSH from Bastion to Private EC2 — Secure tunnel established

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REAL EXPERIENCE

Debugging & Commands

Issue Encountered

SSH returned **permission denied** on first attempt. Root cause: overly permissive key file permissions rejected by SSH.

Fix Applied

```
chmod 400 key.pem
```

SSH Commands Used

```
ssh -i key.pem ec2-user@bastion-public-ip
ssh -i key.pem ec2-user@private-ec2-ip
```

Security Group Rules

Bastion SG

Inbound SSH (22) from `0.0.0.0/0`

Private EC2 SG

Inbound SSH (22) from **Bastion SG only**

Why This Matters

This demonstrates real security thinking: security groups work together to enforce the bastion pattern. Private instances don't know about the internet; they only trust the bastion.

Real hands-on debugging experience — not theoretical knowledge.

Practical, production-level work in action.

This architecture follows a production-grade design where public access, administrative access, and private resource access are clearly separated.

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VALIDATION

Connectivity & Proof

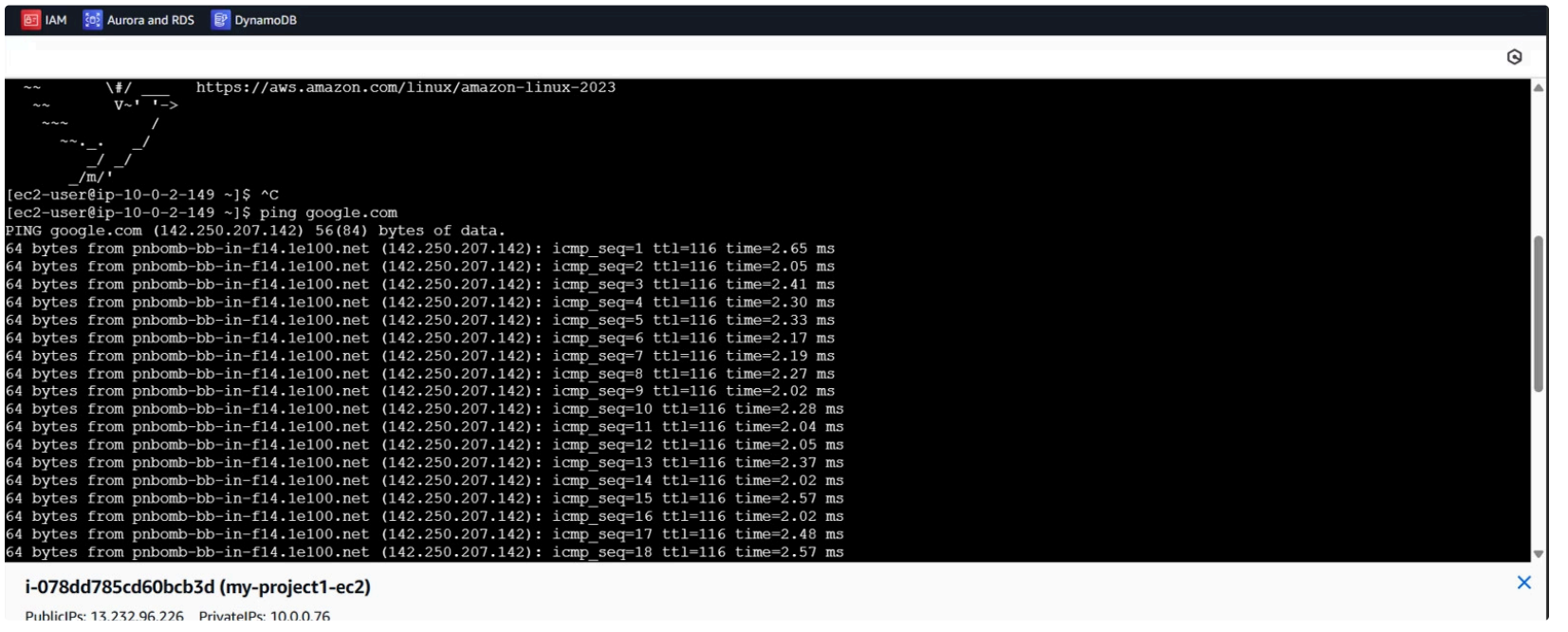
Tests Performed

- `ping google.com` — successful from private EC2
- Outbound traffic confirmed via **NAT Gateway**
- Private instance **not publicly accessible** — verified
- Bastion SSH tunnel to private EC2 — working

What This Proves

- NAT Gateway is functioning correctly
- Route tables are properly configured
- Security groups are enforcing the bastion pattern
- Architecture is production-ready

> Private EC2 can reach the internet, but the internet cannot reach it. This is the core security guarantee of the architecture.



NAT Gateway validation — ping google.com successful from private EC2

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CONCLUSION

Key Learnings

IGW vs NAT Gateway

IGW enables bidirectional internet access; NAT Gateway allows **outbound-only** — critical for private subnet security.

Subnet Design Matters

Public vs private subnet separation, enforced by route tables, is the foundation of network isolation.

Controlled Access

Bastion hosts eliminate direct SSH exposure. Security groups and route tables work together to enforce least privilege.

Production-Ready Skills

Built and validated a secure, scalable AWS VPC — gaining real hands-on cloud engineering experience.

Final System Design Summary

- Public subnet handles external access — internet-facing resources with public IPs
- Private subnet isolates backend services — no public IPs, protected from direct internet exposure
- NAT Gateway ensures outbound internet access only — private instances can reach the internet for updates and API calls
- Bastion Host provides controlled administrative access — single point of audit for all SSH connections

This architecture follows a production-grade design where public access, administrative access, and private resource access are clearly separated.

This project demonstrates real-world AWS networking, secure architecture design, and hands-on debugging experience aligned with production cloud engineering practices. Ready for cloud engineering and DevOps roles.

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